# Model Paper - Database Management Systems

Generated: 2026-03-08 14:04:05

## Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)

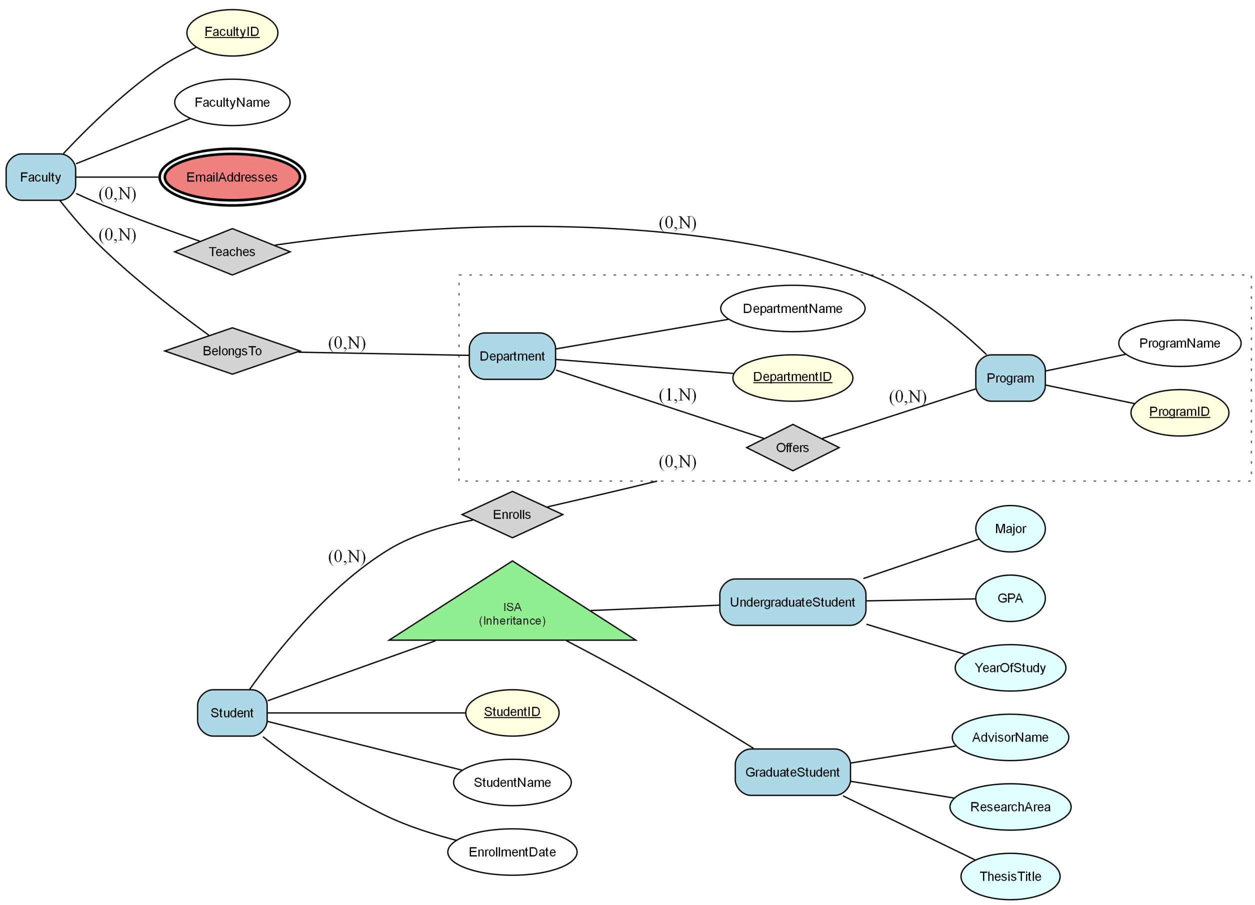
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)

- (0,1) indicates optional participation (zero or one)

- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university setting, there are four main entities: 'Student', 'Course', 'Instructor', and 'Department'. The 'Student' entity has attributes: StudentID, Name, Address, and PhoneNumbers. The 'Course' entity includes CourseID, CourseName, and Credits. Each 'Instructor' has an InstructorID, Name, and DepartmentID, and belongs to a specific 'Department' which has attributes: DepartmentID, DepartmentName, and OfficeLocation. A 'Department' offers multiple 'Courses', and students 'Enroll' in these courses. Each enrollment captures the EnrollmentDate. Furthermore, 'Instructor' can teach multiple 'Courses' within their 'Department', creating a relationship among these entities

**Figure: EER**



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

## Q2 (15 marks)

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies over R: F={{AB->C, B->D, C->E, ADE->B, E->A}}.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

## Q3 (25 marks)

A retail company is looking to establish a database system to manage its inventory, sales, and customer information. The inventory includes products identified by their product IDs, descriptions, prices, and quantities. Sales records link products to customers, capturing transaction details along with timestamps. The company also needs to ensure data integrity and security while allowing multiple user roles such as inventory manager, sales assistant, and customer support.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```
java
                                                                                String s
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) ) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how user roles can be managed in the database system to ensure appropriate access to inventory and sales data. (3 marks)

**e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.**

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provi

fixed serv

## Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Fee (feeId: int, appointmentId: int, amount: real, paymentStatus: varchar(20)) The 'Patient' table stores information about each patient including their unique ID, name, address, and date of birth. The 'Doctor' table contains details about doctors, their unique IDs, names, specialties, and contact information. The 'Appointment' table records the appointments made by patients with doctors, referencing the respective IDs in the 'Patient' and 'Doctor' tables. The 'Fee' table tracks fees associated with appointments, including the payment status. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Fee' table contains relevant information for the database system.

**a) Write SQL Queries to perform the following:**

i. Find the name and phone number of doctors who specialize in 'Cardiology'. (4 marks)

ii. Find

(6 marks)

b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)